# Scaredy Cat

Short manual . free and open source (AGPL-3.0-or-later) . github.com/SpeakingColors

Scaredy Cat is a volume shaper that makes your sound duck, pump or chop in time with the music, without needing a sidechain. It splits the sound into three bands (low, mid, high) and gives each band its own drawn volume curve that repeats every beat, bar or note you choose. Use it for the kick pump in hardstyle and house, trance gates, UK hardcore chops, tremolo, or gentle glue. AMOUNT 0 % is an exact bypass; AMOUNT 100 % is the full depth you drew.

## Quick start

• Pick a preset close to what you want (Hardstyle kick pump, Sidechain house, Trance gate 1/16, UK hardcore chop).
• Press play. The red line in the lanes shows where the cycle is. Set Speed to the note length you want (1/4 = one cycle per beat).
• Draw: drag the round points, double click a lane to add a point, right click a point to delete it. Hold Alt and drag up or down on a lane to bend a segment.
• Raise or lower AMOUNT for the overall depth, and the Low, Mid and High knobs to leave a band alone (0 %) or fully shaped (100 %).
• If the pump is early or late against your kick, move Offset until it lines up.
• Prefer the sound of a hit, not the tempo? Set Trigger to Input hit: every loud hit restarts the envelope.

## How the window works

• Three lanes: HIGH on top, MID in the middle, LOW at the bottom. Each lane is the volume of that band over one cycle; the scale on the left is in dB (top is 0 dB, no change).
• The faint grey bars behind the lanes are your input folded over the cycle, so you can see where the kick hits.
• Under the lanes: shape buttons (Pump, Gate, Swell, Ramp up, Step, Wave, Ramp down, V) and More (trance gate, offbeat pump, random steps, tremolo, reverse, flip, copy to all lanes).
• The tool row above the lanes: select and move, draw a ramp up, ramp down, hold or step by dragging across a lane, delete, select all, and move the selection or the whole lane left and right by one grid step.
• Snap, Grid and Swing sit at the right end of the tool row. Hold Ctrl while dragging to move freely.
• Left: the crossover graph (drag the white lines to move the crossovers), Slope, the three band amounts with Solo buttons, Linked or Unlinked envelopes, Linear phase, and the sidechain options.
• Band spectrum: the sound by frequency at each band's range, dashed before the ducking and filled after. Switch to Level to see each band's peak level in dB.
• Right: the big AMOUNT knob, the output level (OUT) and the biggest gain reduction (GR) in dB, output meters and one gain reduction meter per band.
• Bottom bar: Speed (a note length when Sync is on, Hz when off), Trigger, Drive, Limit, Smooth and Offset. The padlock locks the envelopes so a stray click cannot change them.
• Hover a control for about one second to see an info bubble. Turn bubbles off with the round i button in the header.
• Save stores your settings and all three envelopes as your own preset in the Studio 87 presets folder; they appear in the preset menu after the factory presets.
• Resize the window from its bottom right corner; the 100% button returns to the normal size.

## Controls

## Controls

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| **Power** | Choice: Off / On (default On) | Turns the effect on or off. Off is an exact bypass. |
| **Sync** | Choice: Hz / Sync (default Sync) | Sync follows the host tempo. Hz runs at a fixed speed. |
| **Trigger** | Choice: Repeat / Input hit (default Repeat) | Repeat: the envelope loops with the tempo. Input hit: every loud hit in the input restarts it (no tempo needed). |
| **Slope** | Choice: 12, 24, 48 (default 24) | Steepness of the band split in dB per octave. |
| **Multiband** | Choice: Off / On (default On) | On: three bands with their own envelopes. Off: one envelope for the whole sound. |
| **Linear phase** | Choice: Off / On (default Off) | Splits the bands without phase shift. Adds latency (reported to your host) and a softer low crossover. |
| **Drive type** | Choice: Tube, Tape, Soft (default Tube) | Character of the drive: Tube is warm and even, Tape is gentle, Soft is smooth and round. |
| **Limiter** | Choice: Off / On (default On) | Lookahead limiter after the ducking. |
| **Offset unit** | Choice: MS / SAMPS (default MS) | Offset in milliseconds or in samples. |
| **Envelopes** | Choice: Unlinked / Linked (default Unlinked) | Linked: all three bands use the same envelope (edit one, all follow). |
| **Solo** | Choice: Off, Low, Mid, High (default Off) | Listen to one band only. |
| **Display** | Choice: Input, Sidechain, Both (default Sidechain) | What the dashed spectrum line shows: the input, the sidechain, or both. |
| **Sidechain triggers** | Choice: Off / On (default Off) | The sidechain input (if connected) restarts the envelope on every hit. |
| **Grid** | Choice: 1/4, 1/8, 1/16, 1/32 (default 1/16) | Grid for editing the envelope. |
| **Snap** | Choice: Off / On (default On) | Points snap to the grid while you drag. |
| **Amount** | 0 to 100 %<br>default 100 % | How deep the envelopes cut. 0 % is an exact bypass (the sound passes untouched), 100 % is the full depth you drew. |
| **Low** | 0 to 100 %<br>default 100 % | How much of the low band envelope is applied (0 % leaves the low band alone). |
| **Mid** | 0 to 100 %<br>default 100 % | How much of the mid band envelope is applied. |
| **High** | 0 to 100 %<br>default 100 % | How much of the high band envelope is applied. |
| **Low crossover** | 40 to 400 Hz<br>default 120 Hz | Where the low band ends. Drag the line in the crossover graph or the spectrum. |
| **High crossover** | 800 to 8000 Hz<br>default 2400 Hz | Where the high band starts. |

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| **Speed (Hz)** | 0.05 to 30 Hz<br>default 2 Hz | Speed in cycles per second when Sync is off. |
| **Swing** | 0 to 100 %<br>default 0 % | Delays the second half of every cycle for a swung feel. |
| **Smooth** | 0 to 100 %<br>default 4 % | Rounds the envelope edges so sharp steps do not click. Higher is softer. |
| **Drive** | 0 to 100 %<br>default 0 % | Smooth saturation before the ducking (oversampled). The level stays about the same, the sound gets thicker. |
| **Limit ceiling** | -12 to 0 dB<br>default -0.3 dB | Highest level the limiter lets through. |
| **Offset** | -100 to 100<br>default 0 | Moves the envelope later (positive) or earlier (negative), in ms or samples. Use it to line the pump up with the kick. |
| **Hit threshold** | -60 to 0 dB<br>default -24 dB | How loud a hit must be to restart the envelope in the hit trigger modes. |

## Recommendations

• Kick pump: Speed 1/4, shape Pump, Smooth 3 to 6 %. Keep the High band shallower than the Low band so the hats stay open while the bass ducks.
• Bass only: set Mid and High to 0 % (or use the Bass-only duck preset) so only the low band pumps.
• Trance gate: More, Trance gate x4, Speed 1/4 for sixteenth-note gates; raise Smooth until the clicks disappear.
• No tempo, no kick to follow: Sync off and use Hz, or Trigger Input hit with the threshold just under your kick.
• Smooth steps: sharp steps always click a little; Smooth rounds them. Linear phase is cleaner between bands but adds latency (about 8 to 24 ms at 48 kHz, depending on Slope), which your host compensates.
• Compare with the plugin off (Power) at the same loudness: pumping makes things sound louder or quieter than they are.
• Latency: Scaredy Cat reports its exact latency to your host and updates it when you switch Linear phase or the slope.

## Presets

Default, Hardstyle kick pump, Sidechain house, Trance gate 1/16, Eighth-note pump, UK hardcore chop, Slow tremolo, Gentle glue, Half-time swell, Audio hit duck, Bass-only duck

## Install and use

• Windows: run the installer (.exe). macOS: open the installer (.pkg or .dmg). Linux: install the Flatpak with `flatpak install --user` and the .flatpak file, or copy the .vst3 folder to ~/.vst3.
• Everything works offline. Rescan plugins in your DAW, then look for the plugin under Studio 87.
• Report problems or ideas on the GitHub page of the plugin. Free to use, copy and change under the AGPL-3.0-or-later licence.